



Case Study Solution: Query Practice

SAP Business One 10.0, version for SAP HANA

PUBLIC

INTRODUCTION

These practice exercises are designed to give you hands-on use in creating basic SQL queries for SAP HANA using the built-in query tools in SAP Business One.

You will create the following queries:

1. A report showing a list of customers
2. A report that uses a parameter as selection criteria
3. A report based on multiple tables
4. A report with a running total that can be used with an alert to provide a worklist to a user
5. A query used in a dashboard widget

PREREQUISITE:

1. Use the demo database for SAP Business One 10.0, version for SAP HANA
2. Credentials: User code: **manager**

Use either of the SAP Business One query tools: **Query Generator** or **Query Wizard**. Note: the solutions are shown for the **Query Generator** only. All queries shown here use the HANA SQL syntax.

Important note: The reports included in this case study show data from the UK localization database. The data you will see will obviously be different depending on your localization and the date you run the queries.

Task 1

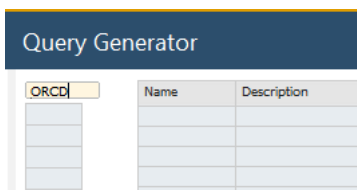
Create a Customer List Report

This report displays a list of customers and balances from the **OCRD** Business Partners table.

Create the query

Choose *Tools* → *Queries* → *Query Generator*.

Type **OCRD** in the highlighted field at the top left of the window, then press **Tab**.



Select fields for the report by double-clicking in the list:

- CardCode
- CardName
- Address
- City (Bill-to city)
- ZipCode
- Balance
- CntctPrsn

Note: When using a HANA database, multi-case field names must be enclosed in double quote marks in the query.

Tip: Sort the Fields

To make it easier to select fields, you can sort the list of fields alphabetically by double-clicking in the Name column header.

Name ▲	Description
Address	Bill to
Address2	Ship To
AgentCode	Agent Code
AgrNo	Agreement No.
AltBaseEnt	Alteration Base Entry
AltBaseTyp	Alteration Base Type
AnnInvDecR	Annual Inv. Declaration Ref.

Add a condition to the query

Since the OCRD table holds records for leads and vendors as well as customers, you need to add a filter for **customer** master records:

- Click in the *Where* area on the right.
- Double-click to select the CardType field.
- Type = 'C' to complete the *Where* clause (use the single quote character). The *Where* clause should now read: **T0."CardType" = 'C'**

Choose *Execute*.

Query Generator

Table: OCRD

Name	Description
CardCode	BP Code
CardName	BP Name
CardType	BP Type
GroupCode	Group Code
CmpPrivate	Business Partner Type
Address	Bill-to Street
ZipCode	Bill-to Zip Code
MailAddress	Ship-to Street
MailZipCod	Ship-to Zip Code
Phone1	Telephone 1
Phone2	Telephone 2
Fax	Fax Number
CntctPrsn	Contact Person
Notes	Remarks
Balance	Account Balance
ChecksBal	Open Checks Balance
DNotesBal	Open Deliveries/GRPO Balar
OrdersBal	Open Orders Balance

Select: T0."CardCode", T0."CardName", T0."Address", T0."Balance", T0."ZipCode", T0."City", T0."CntctPrsn"

From: OCRD T0

Where: T0."CardType" = 'C'

Sort By:

Group By:

Buttons: Execute, Close, Conditions

Fine Tune the Results

- You can sort the results according to any column by **double-clicking** in the header field of the column. You can alternately add a sort to the query by entering a field in the *Sort By* area of the Query Generator window.
- To include a total of the account balances, press **Ctrl and click** in the header field of the Account Balance column. The total will appear at the bottom of the column, as shown below.

Query Preview

```
SELECT T0."CardCode", T0."CardName", T0."Address", T0."Balance", T0."ZipCode", T0."City", T0."CntctPrsn"
FROM OCRD T0 WHERE T0."CardType" ='C' ORDER BY T0."Balance"
```

Ctrl+Click

#	BP Code	BP Name	Bill-to Street	Account Balance	Bill-to Zip Code	Bill-to City	Contact Person
1	C60000	SG Electronics	45th Street	-27,077.56	10010	New York	Eric Alexander
2	C23900	Parameter Technology	908 Darby Road	-15,356.78	B1 6TR	Birmingham	Daniel Brown
3	C40000	Earthshaker Corporation	94 Bolton Road	-1,653.03	NW12 8HG	London	Bob McKensly
4	C50000	ADA Technologies	FluggeStrasse 6	0.00	22303	Hamburg	Mary Brown
5	C1234	VIP		0.00			
6	C42000	Mashina Corporation	400 Range Road	36.00	M3 7FD	Glasgow	Anthony Smith
7	C99999	One Time Customer		463.09			
8	C25000	Star Company	Plynaresnska	975.18	821 09	Bratislava	Martina Basalykova
9	C30000	Microchips	45th Street	1,392.00	N1 8GH	Leeds	Judy Brown
10	C99998	Web Customer		1,596.00			
11	C70000	Aquent Systems	Hollywood Ave	2,325.72	95112	San Jose	Troy Brown
12	C26000	River Ltd.	United Rd	5,094.00	M16 0RA	Manchester	Lewis Beerman
13	C20799	34 Technologies		13,200.00			
14	C20000	Maxi-Teq	Hendon Way	13,866.20	NW2 7YT	London	Norm Thompson
15	C80000	Pivotal Solutions	8th Avenue	50,217.77		New York	
				45,078.59			

Execute Cancel Reverse Table Copy Data Save Save As Open

Save the query for reuse

In the **Query Preview** window, choose **Save**.

In the **Save Query** window, choose **Manage Categories**.

Enter a new category called **Sales**.

Choose **Add\Update**.

Save Query

Query Name

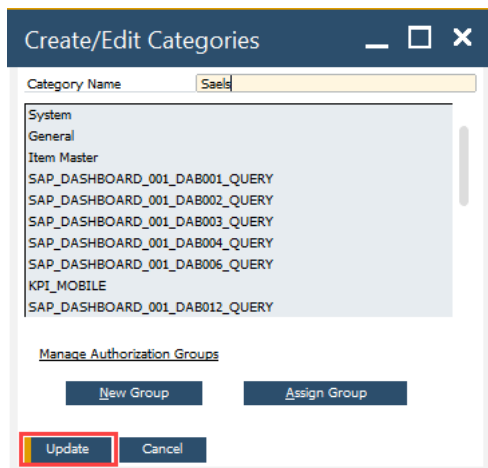
Query Category

Manage Categories

General

Item Master

Save Cancel Schedule New Stored Procedure Remove



The 'Create/Edit Categories' dialog box shows the 'Category Name' field with 'Sales' entered. Below the list of categories, the 'Update' button is highlighted with a red rectangle.

Category Name: Sales

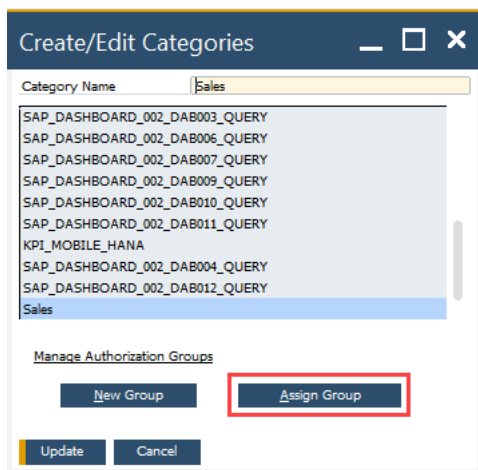
System
General
Item Master
SAP_DASHBOARD_001_DAB001_QUERY
SAP_DASHBOARD_001_DAB002_QUERY
SAP_DASHBOARD_001_DAB003_QUERY
SAP_DASHBOARD_001_DAB004_QUERY
SAP_DASHBOARD_001_DAB006_QUERY
KPI_MOBILE
SAP_DASHBOARD_001_DAB012_QUERY

Manage Authorization Groups

New Group Assign Group

Update Cancel

Select the newly created Sales category and choose the *Assign Group* button to assign the category to the *Saved Queries – Group No.1* authorization group.



The 'Create/Edit Categories' dialog box shows the 'Category Name' field with 'Sales' entered. The 'Sales' category is selected in the list. The 'Assign Group' button is highlighted with a red rectangle.

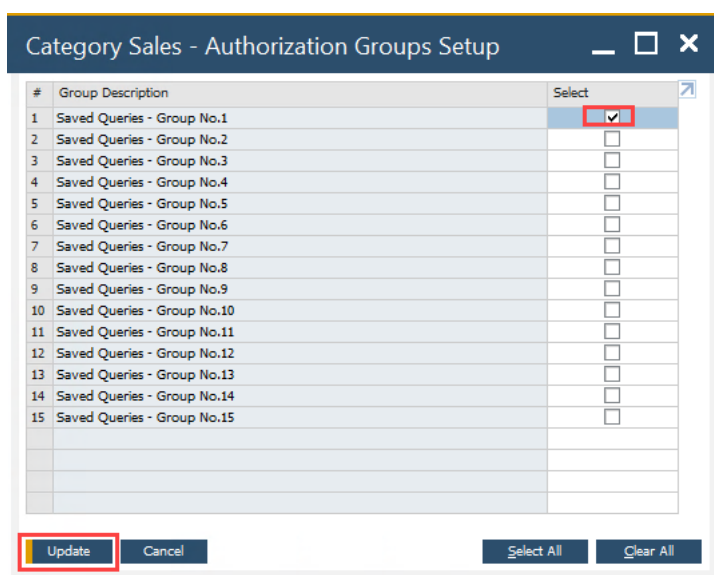
Category Name: Sales

SAP_DASHBOARD_002_DAB003_QUERY
SAP_DASHBOARD_002_DAB006_QUERY
SAP_DASHBOARD_002_DAB007_QUERY
SAP_DASHBOARD_002_DAB009_QUERY
SAP_DASHBOARD_002_DAB010_QUERY
SAP_DASHBOARD_002_DAB011_QUERY
KPI_MOBILE_HANA
SAP_DASHBOARD_002_DAB004_QUERY
SAP_DASHBOARD_002_DAB012_QUERY
Sales

Manage Authorization Groups

New Group Assign Group

Update Cancel



The 'Category Sales - Authorization Groups Setup' dialog box shows a table with 15 rows. The first row, 'Saved Queries - Group No.1', has its 'Select' checkbox checked. The 'Update' button is highlighted with a red rectangle.

#	Group Description	Select
1	Saved Queries - Group No.1	☒
2	Saved Queries - Group No.2	☐
3	Saved Queries - Group No.3	☐
4	Saved Queries - Group No.4	☐
5	Saved Queries - Group No.5	☐
6	Saved Queries - Group No.6	☐
7	Saved Queries - Group No.7	☐
8	Saved Queries - Group No.8	☐
9	Saved Queries - Group No.9	☐
10	Saved Queries - Group No.10	☐
11	Saved Queries - Group No.11	☐
12	Saved Queries - Group No.12	☐
13	Saved Queries - Group No.13	☐
14	Saved Queries - Group No.14	☐
15	Saved Queries - Group No.15	☐

Update Cancel Select All Clear All

Choose *Update* then *OK*.

Back in the **Save Query** window, select the *Sales* category.

Enter **Customer Balance Report** in the *Query Name* field. After that choose *Save*.

The screenshot shows the 'Save Query' dialog box with the following details:

- Query Name:** Customer Balance Report
- Query Category:** Sales (selected from a dropdown menu)
- Buttons:** Save, Cancel, Schedule, New Stored Procedure, Remove

Run the saved query

Choose *Tools* → *Queries* → *User Queries* → *Sales* → *Customer Balance Report*.

Task 2

Create a Report with a Parameter

This report displays a list of customer invoices posted after a certain date. The date is entered as a parameter when the query is run. The report uses the OINV invoice table.

Find Field Names

To find out the field names for the report, use system information:

- Open up a blank A/R invoice document and toggle on *View > System Information*.
- Hold your mouse over the following fields and write down the database field name that shows in the system information area:

Name in Document	Database Field Name
<i>No.</i>	
<i>Customer</i>	
<i>Name</i>	
<i>Posting Date</i>	
<i>Total</i>	DocTotal

Note: When you hold your mouse over the Total field, the field name does not display in system information. This is because this field holds the currency symbol as well as the amount. The database field name is *DocTotal*.

Create the Query

In the Query Generator window, choose the **X** button to clear out the previous table selection. If you closed the Query Generator window, re-open it using the path **Tools → Queries → Query Generator**.

Type **OINV** in the *Table* field and press **Tab**.

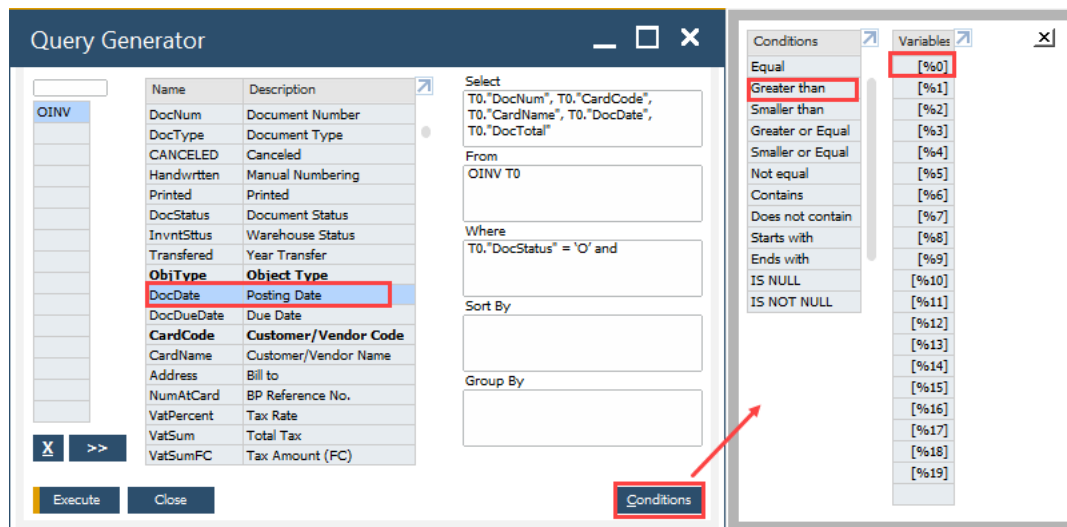
Select the fields from the OINV table that you identified using system information.

In the Where clause, add a filter for open documents (DocStatus = 'O'). The *Where* clause should read:

T0."DocStatus" = 'O'

In the *Where* clause, type the word 'and' then select the document date field (**DocDate**). Then press the *Conditions* button to open the side window:

- Double-click the *Greater than* condition
- Double-click the first variable [%0].

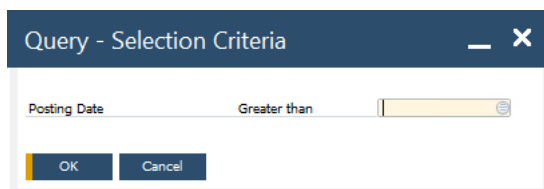


The *Where* clause should now read:

T0."DocStatus" = 'O' and T0."Docdate" > [%0]

Run the Query

Choose *Execute*. A popup window will appear.



Choose the selection list icon in the popup window then select a date from the list of results. Note: if you are using the demo database the invoices you see may be very old, unless you have recently added invoices.

Choose OK to run the query.

The system displays the result. Notice the orange drill-down arrows to the customer master data.

To display the total for the invoices, choose Ctrl and double-click the *Document Total* column heading.

[illegible]

Save this query with the name **Invoice List** in the *Sales* category.

Task 3

Create a Report from Multiple Tables

This query will display a list of open sales quotations summarized by customer and grouped by sales employee. The query will use two tables:

- Sales Quotation (OQUT)
- Sales Employee table (OSLP)

The inner join will be provided for you by the query tool.

Create the query

Clear the previous table from the query window.

Enter each table name in upper left box and press **Tab** each time. Notice the inner join is made for you in the query generator window.

Select the OSLP table and then select the **SlpName** field.

Select the OQUT table and then select the **CardCode** and **CardName** fields.

Calculate the total value of the sales quotations using the SUM function and provide a heading in the report:

SUM(T0."DocTotal") as "Total Value"

Count the number of sales quotations for each customer using the COUNT function and provide a heading:

COUNT(T0."DocNum") as "No of Documents"

Important: Make sure you do not forget the closing parenthesis for the SUM and COUNT functions.

Query Generator

Name	Description
CANCELED	Canceled
CardCode	Customer/Vendor Code
CardName	Customer/Vendor Name
CashDiscFC	Cash Discount (FC)
CashDiscnt	Cash Discount
CashDiscPr	Cash Discount Percentage
CashDiscSC	Cash Discount (SC)
CCDEntry	CCD Abs. Entry
CdcOffset	Cash Discount Offset
CEECFlag	A/P Correction Invoice
CertifNum	Certification Number
CertNum	Certificate Number
CheckDigit	Control Digit
Checker	Checker
CIG	Contract Code Identificat
ClosingOpt	Closing Option
ClsDate	Document Closing Date
CntctCode	Contact Person
CntrlBnk	Central Bank Indicator
Comments	Remarks
ComTrade	Commission Trade
ComTradeRt	Commission Trade Return
Confirmed	Confirmed

>>

Select
 T1."SlpName", T0."CardCode",
 T0."CardName", SUM(T0."DocTotal")
 as "Total Value",
 COUNT(T0."DocNum") as "No of
 Documents"
 From
 OQUT T0 INNER JOIN OSLP T1 ON
 T0."SlpCode" = T1."SlpCode"
 Where
 T0."DocStatus"='O'
 Sort By
 T1."SlpName"
 Group By
 T1."SlpName", T0."CardCode",
 T0."CardName"

Enter the following condition in the *Where* clause so that only open quotations are used:

T0."DocStatus" = 'O'

Click in the *Sort by* area and select the **SlpName** field.

Click in the *Group by* area and select **SlpName**, **CardCode** and **CardName** fields so that the results are grouped by sales employee and customer.

Run the Query

Choose Execute.

The results show for each sales employee a count of open sales quotations for each customer and the total value of the quotations. For example:

[illegible]

Save the Query.

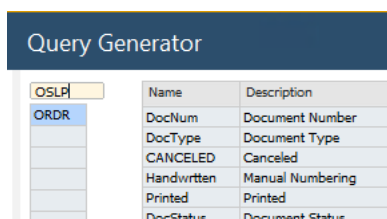
Task 4

Create a Report as a Worklist for a User

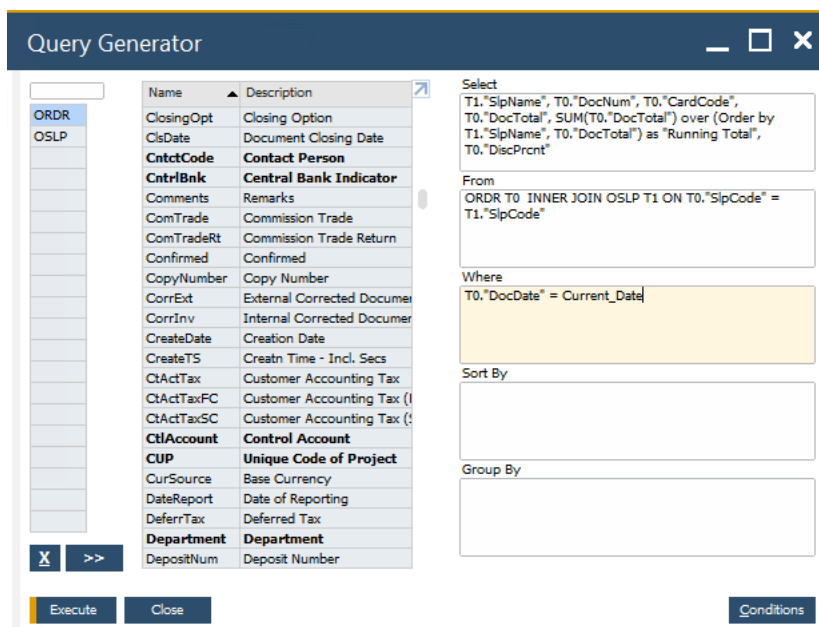
This report displays all the sales orders posted for today, organized by sales employee name. The report uses the ORDR table and the OSLP table. The report is saved then can be later used with an alert to provide a daily worklist for a user.

Preparation: Create 2-3 sales orders for customers, with today's posting date. Select a discount % in each order.

In the Query Generator, select the ORDR table, press Tab, then select the OSLP table and press tab.



Enter the SQL as shown below in the Select area. Select each table before selecting the fields. Note the query uses the SUM expression to total the sales orders combined with the OVER clause to maintain a running total by sales employee. Enter the Where clause as shown below.



Execute the query

[illegible]

Save the query in the *Sales* category with the name **Today's Sales Orders**.

INFORMATION: The report can be scheduled to run daily and sent to a user using the alert mechanism. This will be covered in the case study for alerts.

Create a Query for a Dashboard Widget

Set up a count widget that counts the number of deliveries created each day.

Create the query

The query should select document numbers from the ODLN table where the document date (posting date) is the current date.

As shown below:

The screenshot shows the SAP Query Generator window. On the left, the 'ODLN' table is selected. The central pane displays the table's fields, with 'DocDate' (Posting Date) highlighted. The right pane shows the query configuration: 'Select' is set to 'T0."DocDate"', 'From' is 'ODLN T0', and 'Where' is 'T0."DocDate" = CURRENT_DATE'. The 'Sort By' and 'Group By' fields are empty. At the bottom, there are buttons for 'Execute', 'Close', and 'Conditions'.

Save and name the query **Deliveries_Today**.

Test the Query

Create a few deliveries (with the current date) so that you can test your query.

Sales – A/R → Delivery

Here is an example delivery:

[illegible]

After creating a few deliveries, return to your query and choose *Execute*.

The results show a list of delivery documents for today's date. For example:

[illegible]

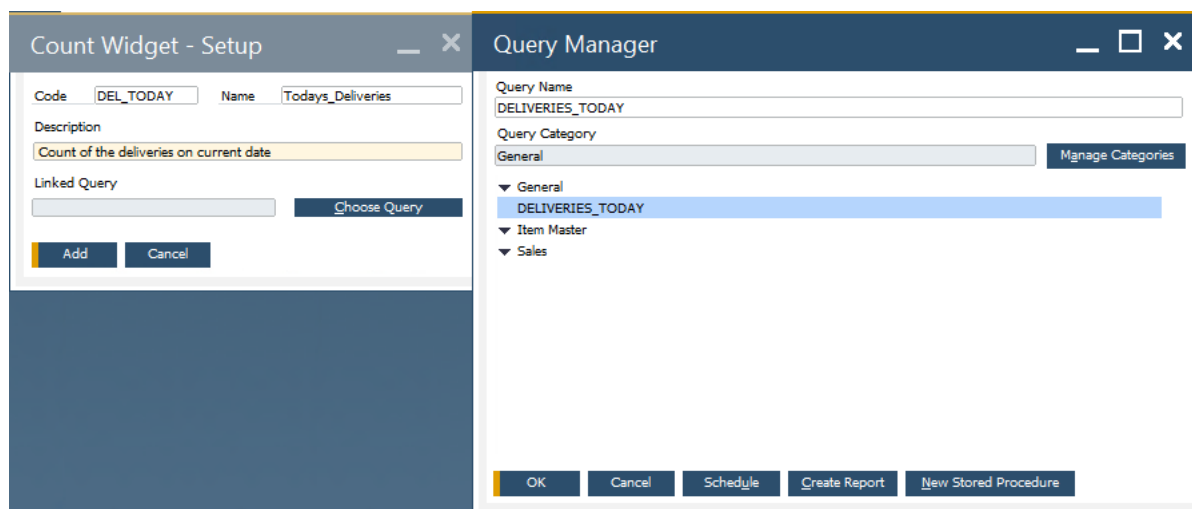
Set up the Count Widget

Tools > Cockpit → Count Widget Setup

Switch to add mode .

Give the count widget a code, name and description. The name will appear on the widget so make it short and precise. For example, name it **Today's Deliveries**.

Link the count widget to the query you just created: Select *Choose Query*.
The *Query Manager* window opens. Choose your query and select *OK*.



When your query shows in the *Count Widget – Setup* window, Choose *Add*.

Add the Count Widget to your cockpit

Choose the *Pencil* icon to make changes to your cockpit.

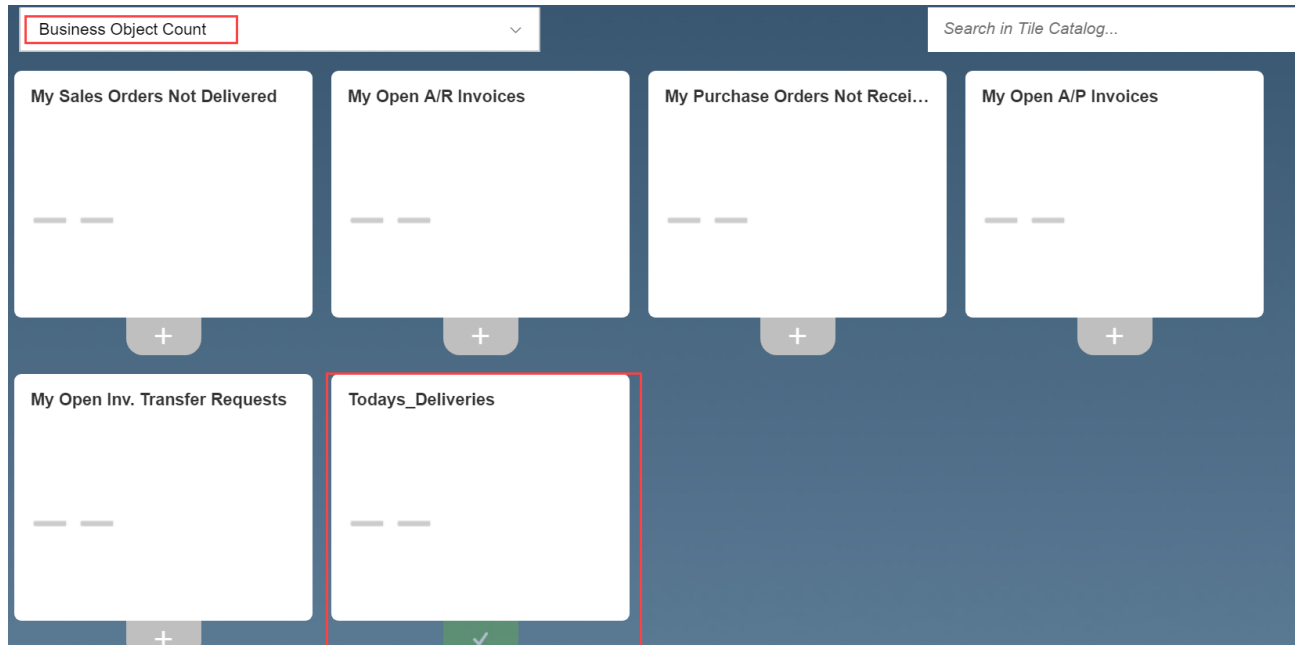


Use the + symbol to open the Widget Gallery.



Choose *Business Object Count* in the dropdown box.

To choose the widget, click on the Plus sign below the widget. It will change to a green checkmark.



Choose the arrow in the top left to return to the cockpit.

Use the icon shaped like a checkmark to save the changes to the cockpit.



Test the Count Widget

You can test the count widget by adding more deliveries. You can wait until the cockpit picks up the new deliveries or chose *Refresh* to see the count updated.



If you click on the number displayed, a window will open with the query results.



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